Antariksha Hanmant Dhanure

+1 (716)2928088 | adhanure@buffalo.edu | Github | Portfolio | LinkedIn

EDUCATION

State University of New York at Buffalo, USA **Dec 2026**
Masters in computer science and engineering 3.8/4
Rajiv Gandhi Institute of Technology, University of Mumbai, India **May 2025**
Bachelor of Engineering in Artificial Intelligence and Data Science 8/10

COMPUTING PROFICIENCY

Languages: Python, SQL, R, React.js, JavaScript, JSON, HTML, CSS.

Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Hugging, XGBoost, Pandas, NumPy, SciPy, Matplotlib, Seaborn, Optuna, Large Language Model, Explainable AI, CNNs, GNNs, Natural Language Processing.
Tools: Jupyter Notebook, Google Collab, GitHub, Streamlit, Flask, AWS, Tableau, Power BI, VS Code, Spyder, Docker, Linux/Unix Cursor AI.
Databases: MySQL, PostgreSQL, MongoDB, SQLite, AWS Cloud Storage

WORK EXPERIENCE

Graduate Research Assistant, SUNY at Buffalo, New York **Oct 2025 to Jan 2026**
- Spearheaded research on anomaly and fault detection in medical **time-series** data from **Continuous Glucose Monitoring (CGM)** systems used in artificial pancreas platforms.
- Built and evaluated unsupervised machine learning models enabling early detection of sensor faults within 3–6 hours of onset.
- Engineered time-series feature pipelines using rolling statistics, signal derivatives, and trend analysis, improving anomaly separability **and reducing false positives by 25–30%** after threshold tuning.
- Validated models on simulated and real-world **CGM datasets**, achieving high recall on fault events while maintaining low false-alarm rates for safety-critical monitoring

Data Analyst Intern at AB Infinity Soft Solutions, Hyderabad, India **May 2023 to July 2023**
- Analyzed web traffic, **SEO** performance, and social media engagement data **using SQL,** Power BI, and Google Analytics, supporting data-driven marketing decisions across campaigns.
- Conducted keyword research and analysis, **contributing to 20% improvement in organic reach** for targeted content initiatives.
- Performed sentiment analysis on social media data, identifying user trends and improving campaign messaging alignment.
- Built reusable dashboards and automated reports, **reducing manual reporting time by 30%** for recurring analyses.

PROJECTS

AI-Based Deskstop Assistant
- Designed and built an end-to-end audio classification system using **MFCC** feature extraction and **CNNs in PyTorch** to address poor voice recognition in noisy environments.
- Applied signal preprocessing techniques, improving robustness across real-world acoustic conditions.
- Trained and **optimized CNN architectures, achieving 25–35%** improvement in speech command recognition accuracy over baseline models in noisy settings

Predictive Modeling for Loan Approval
- Built supervised **ML models** to automate loan approvals, reducing manual review effort by 40% overrule-based screening.
- Applied feature engineering and model evaluation, improving F1-score across classifiers.
- Compared **Logistic Regression, Random Forest, and Gradient Boosting** models to identify optimal trade-offs between accuracy (+18%), fairness metrics, and interpretability

Fashion Recommendation System
- Extracted visual embeddings from fashion images using **CNN-based feature extraction**.
- Combined deep visual features with **K-Nearest Neighbors (KNN)** similarity search to generate recommendations.
- Designed a hybrid pipeline integrating image features with user interaction data.
- **Improved recommendation relevance by 20–25%** compared to non-visual, popularity-based baselines.

Portfolio Website
- **Leveraged Cursor AI** (AI-assisted coding) to accelerate component development, **reducing development time by ~30–40%.**
- Implemented **modular React** components and clean UI layouts, improving page load performance and maintainability.
- Deployed the portfolio as a production-ready web application, **enabling centralized access to GitHub**, resume, and projects.

PUBLICATIONS

- Study on Predictive Modeling for Loan Approval. International Research Journal on Advanced Engineering and Management (IRJAEM). DOI: 10.47392/IRJAEM.202
- TRUSTNOVA: A Predictive Machine Learning Model for Loan Approval. International Journal for Research in Applied Science and Engineering Technology (IJRASET). DOI: 10.22214/ijraset.2025.68388